Assembly & Machine Language Lab

Assignment 3

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2nd Year 1st Semester

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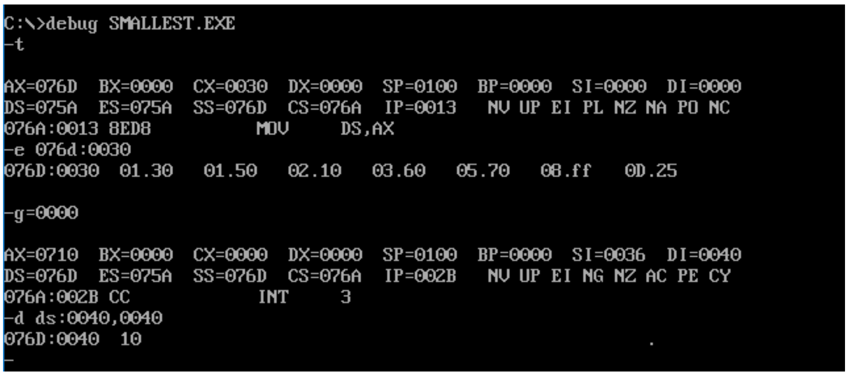
1. Write an Assembly Language Program to find the smallest number from a series of seven data bytes stored from DS: 0030H. Store the smallest number in DS: 0040H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov cx,0007h
mov di,0040h
mov al,[si]
l1:     inc si
        cmp al,[si]
        jc l2
        mov al,[si]
l2:     loop l1
mov [di],al
int 03h
mov ah,4ch
int 21h
main endp
end main
```

Output:

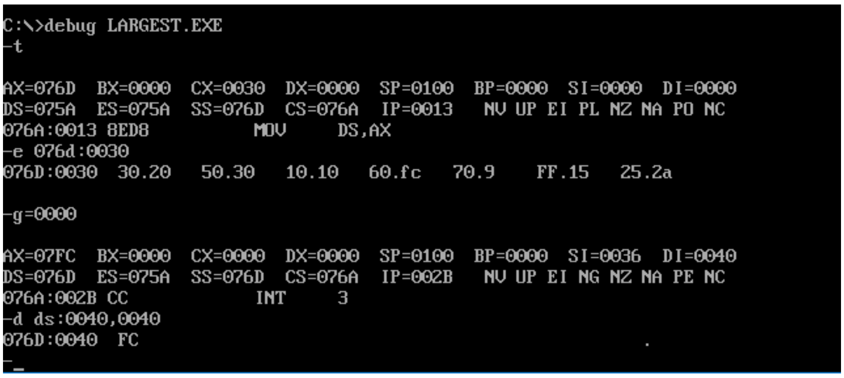


```
C:\>debug SMALLEST.EXE
-t

AX=076D  BX=0000  CX=0030  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0013   NV UP EI PL NZ NA PO NC
076A:0013 8ED8          MOV     DS,AX
-e 076d:0030
076D:0030  01.30   01.50   02.10   03.60   05.70   08.ff   0D.25

-g=0000

AX=0710  BX=0000  CX=0000  DX=0000  SP=0100  BP=0000  SI=0036  DI=0040
DS=076D  ES=075A  SS=076D  CS=076A  IP=002B   NV UP EI NG NZ AC PE CY
076A:002B CC            INT     3
-d ds:0040,0040
076D:0040  10                                            .
-
```

2. Write an Assembly Language Program to find the largest number from a series of 7 sixteen-bit numbers stored from DS: 0030H. Store the largest number in DS: 0040H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov cx,0007h
mov di,0040h
mov al,[si]
l1:     inc si
        cmp al,[si]
        jnc l2
        mov al,[si]
l2:     loop l1
mov [di],al
int 03h
mov ah,4ch
int 21h
main endp
end main
```

Output:



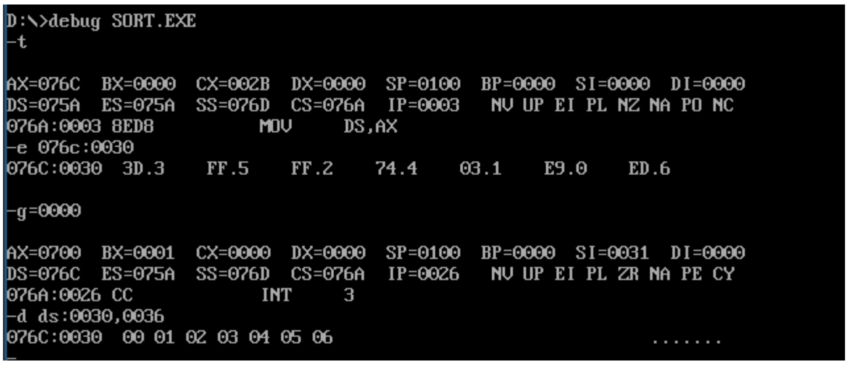
```
C:\>debug LARGEST.EXE
-t

AX=076D  BX=0000  CX=0030  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0013   NV UP EI PL NZ NA PO NC
076A:0013 8ED8          MOV     DS,AX
-e 076d:0030
076D:0030  30.20   50.30   10.10   60.fc   70.9    FF.15   25.2a

-g=0000

AX=07FC  BX=0000  CX=0000  DX=0000  SP=0100  BP=0000  SI=0036  DI=0040
DS=076D  ES=075A  SS=076D  CS=076A  IP=002B   NV UP EI NG NZ NA PE NC
076A:002B CC            INT     3
-d ds:0040,0040
076D:0040  FC                                                .
-_
```

3. Write an Assembly Language Program to arrange a series of 7 data bytes stored from DS: 0030H in ascending order.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
    mov ax, @data
    mov ds, ax
    mov si, 0030h
    mov cx, 0006h
outer_loop:
    mov si, 0030h
    mov dx, cx
```

```
inner_loop:
    mov al, [si]
    mov bl, [si+1]
    cmp al, bl
    jc no_swap
    xchg al, bl
    mov [si], al
    mov [si+1], bl
no_swap:
    inc si
    dec dx
    jnz inner_loop
    loop outer_loop
    int 3h
    mov ah, 4ch
    int 21h
main endp
end main
```

Output:



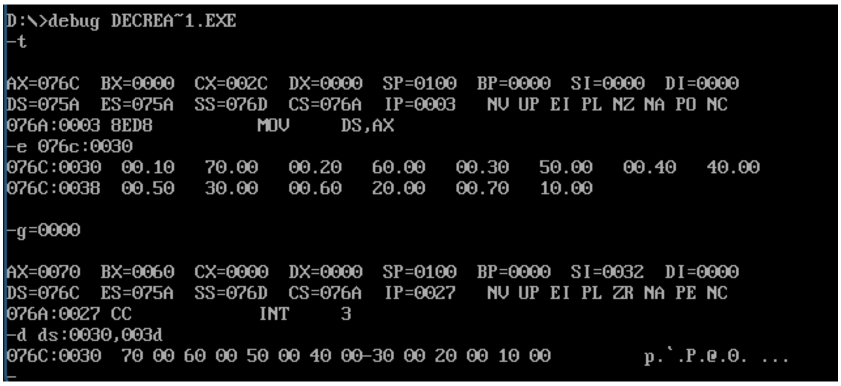
```
D:\>debug SORT.EXE
-t

AX=076C  BX=0000  CX=002B  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0003   NV UP EI PL NZ NA PO NC
076A:0003 8ED8          MOV     DS,AX
-e 076c:0030
076C:0030  3D.3    FF.5    FF.2    74.4    03.1    E9.0    ED.6

-g=0000

AX=0700  BX=0001  CX=0000  DX=0000  SP=0100  BP=0000  SI=0031  DI=0000
DS=076C  ES=075A  SS=076D  CS=076A  IP=0026   NV UP EI PL ZR NA PE CY
076A:0026 CC            INT     3
-d ds:0030,0036
076C:0030  00 01 02 03 04 05 06                            .......
```

4. Write an Assembly Language Program to arrange a series of 7 sixteen-bits data stored from DS: 0030H in descending order.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
    mov ax, @data
    mov ds, ax
    mov si, 0030h
    mov cx, 6
outer_loop:
    mov si, 0030h
    mov dx, cx
inner_loop:
    mov ax, [si]
    mov bx, [si+2]
    cmp ax, bx
    jnc no_swap
    xchg ax, bx
    mov [si], ax
    mov [si+2], bx
```

```
no_swap:
    add si, 2
    dec dx
    jnz inner_loop
    loop outer_loop
    int 3h
    mov ah, 4ch
    int 21h
main endp
end main
```

Output:



```
D:\>debug DECREA~1.EXE
-t

AX=076C  BX=0000  CX=002C  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0003   NV UP EI PL NZ NA PO NC
076A:0003 8ED8          MOV     DS,AX
-e 076c:0030
076C:0030  00.10   70.00   00.20   60.00   00.30   50.00   00.40   40.00
076C:0038  00.50   30.00   00.60   20.00   00.70   10.00

-g=0000

AX=0070  BX=0060  CX=0000  DX=0000  SP=0100  BP=0000  SI=0032  DI=0000
DS=076C  ES=075A  SS=076D  CS=076A  IP=0027   NV UP EI PL ZR NA PE NC
076A:0027 CC            INT     3
-d ds:0030,003d
076C:0030  70 00 60 00 50 00 40 00-30 00 20 00 10 00        p.`.P.@.0. ...
```

5. Write an Assembly Language program to find the square of a number stored in DS:0030H using LOOK-UP table. Assume that the LOOK-UP table is stored from DS:0040H that contains the square of the numbers 0 to 9. Store the square value in DS:0050H.

DS:0100H	00
DS:0101H	01
DS:0102H	04
DS:0103H	09
DS:0104H	16
DS:0105H	25
DS:0106H	36
DS:0107H	49
DS:0108H	64
DS:0109H	81

```
dosseg
.model small
.stack 100h
.data
.code
main proc
    mov ax,@data
    mov ds,ax
    mov es,ax
```

```
    mov bx,0040h
    mov si,0030h
    mov al,[si]
    xlat
    mov di,0050h
    mov [di],al
    int 3h
    mov ah,4ch
    int 21h
main endp
end main
```

Output:

```
D:\>debug LOOKUP.EXE
-t

AX=076C  BX=0000  CX=002A  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0013   NV UP EI PL NZ NA PO NC
076A:0013 8ED8          MOV     DS,AX
-e 076c:0040
076C:0040  E4.0    40.1    50.4    8B.9    C3.10   8C.19   C2.24   05.31
076C:0048  0C.40   00.51

-e 076c:0030
076C:0030  3D.7

-g=0000

AX=0731  BX=0040  CX=002A  DX=0000  SP=0100  BP=0000  SI=0030  DI=0050
DS=076C  ES=076C  SS=076D  CS=076A  IP=0025   NV UP EI PL NZ NA PE CY
076A:0025 CC            INT     3
-d ds:0050,0050
076C:0050  31                                                1
-
```